Yiran Lei

Ph.D. Student in Computer Science, Carnegie Mellon University

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I measure GPU systems to understand what slows communication and where energy is wasted, then use those insights to design more efficient systems.

Research Highlights

Traffic scheduling across scale-up and scale-out GPU networks NSDI 2026

Mixture-of-experts models generate skewed, rapidly changing all-to-all traffic. We use fast scale-up links to rebalance traffic within servers, then schedule scale-out transfers to avoid incast.

Energy cost of idle GPUs arXiv 2026

A GPU with a program loaded can draw more power than a fully idle GPU, even when it is doing no work. Over a month on a CMU cluster, this execution-idle state accounted for ∼**10%** of the GPU energy.

Education

Carnegie Mellon University Ph.D. in Computer Science 2023–Present
Advisor Justine Sherry

Tsinghua University M.E. in Computer Science 2020–2023
Advisor Mingwei Xu

Tsinghua University B.E. in Software Engineering 2016–2020

Publications

FAST: An Efficient Scheduler for All-to-All GPU Communication NSDI 2026

Yiran Lei, Dongjoo Lee, Liangyu Zhao, Daniar Kurniawan, Chanmyeong Kim, Heetaek Jeong, Changsu Kim, Hyeonseong Choi, Liangcheng Yu, Arvind Krishnamurthy, Justine Sherry, and Eriko Nurvitadhi.

SwitchNIC: An Hybrid Architecture for Network Functions with Fast and Consistent Shared State CoNEXT 2025

Yiran Lei, Francisco Pereira, Arvind Krishnamurthy, and Justine Sherry.

PrintQueue: Performance Diagnosis via Queue Measurement in the Data Plane SIGCOMM 2022

Yiran Lei, Liangcheng Yu, Vincent Liu, and Mingwei Xu.

DOVE: Diagnosis-driven SLO Violation Detection ICNP 2021

Yiran Lei, Yu Zhou, Yunsenxiao Lin, Mingwei Xu, and Yangyang Wang.

Preprint

The Energy Cost of Execution-Idle in GPU Clusters arXiv 2026

Yiran Lei, Jared Fernandez, Vasilis Kypriotis, Dimitrios Skarlatos, Emma Strubell, Justine Sherry, and Daniel Vosler.

Research Experience

Microsoft Research Research Intern — May–Aug 2025
Designed programmable congestion control for large-scale AI networks. — Mentor Michael Papamichael

MangoBoost Technical Intern — May 2024–Feb 2025
Co-developed FAST for MoE training across two-tier GPU fabrics. — Mentor Eriko Nurvitadhi

University of Washington Visiting Research Intern — Sep 2022–May 2023
Developed SwitchNIC's shared-state mechanism for network functions split across a programmable switch and SmartNICs. — Advisor Arvind Krishnamurthy

University of Pennsylvania Research Intern — Sep 2021–Aug 2022
Built PrintQueue to diagnose congestion from switch queue measurements. — Advisor Vincent Liu

UCLA Visiting Research Intern — Jun–Aug 2019
Built a home IoT system using Named Data Networking. — Advisor Lixia Zhang

Selected Talks

The Energy Cost of Execution-Idle in GPU Clusters — Jun 2026
Parallel Data Laboratory Seminar, Carnegie Mellon University

FAST: An Efficient Scheduler for All-to-All GPU Communication — Apr 2026
Distributed Systems Laboratory Seminar, University of Pennsylvania

Teaching

Carnegie Mellon University Teaching Assistant — Fall 2025, Fall 2026
Computer Networks and the Internet (15-441/641). Led recitations and gave guest lectures on TCP, HTTP, and data center networking.

Tsinghua University Teaching Assistant — Fall 2021
The Principle of Computer Network (40240513). Gave guest lectures on IPv6.

Selected Honors

CMU SCS Fellowship in Computer Science — 2026

China National Scholarship — 2022

Fellowship for Comprehensive Excellence (Second Class), Tsinghua — 2021

Technical Expertise

GPU systems and measurement CUDA, HIP, NVSHMEM, NCCL/RCCL, NVML, DCGM

Programmable networks P4/Tofino, DPDK, SmartNICs (Stingray, BlueField-3)

Programming C/C++, Python